American Computer Science League

2023-2024 ● Contest 2: ACSL Book ● Junior Division

PROBLEM: A book cipher is a method of encoding and decoding a message where the sender and the receiver both use the same text. The sender's message is encoded by replacing each alphanumeric character in the message to be sent with the location of that character in the text using an "*s.w.c*" format representing the number of the sentence in the text (*s*), the number of the word in that sentence (*w*), and the number of the character in that word (*c*), all indexed starting with 1. The receiver decodes the encoded message by locating the characters in the text specified by each "*s.w.c*" string.

In this program, you will be given the text and the encoded message to be decoded. The text will contain alphanumeric characters, spaces, and periods using the following restrictions:

- A word contains only alphanumeric characters
- All words are separated by a single space
- Sentences are terminated with a period and separated by two spaces

The encoded message will contain multiple "*s.w.c*" strings, each separated by a single space. Create the decoded message by finding the characters in the text. If an "*s.w.c*" string doesn't refer to a character in the text, use a space in the decoded message.

EXAMPLE:

Input:

```
ACSL is an international computer science competition among more than
300 schools.  With countrywide and worldwide participants it became
the American Computer Science League.  It has been in continuous
existence since 1978.  Each yearly competition consists of 4 regular
season contests.  All students at each school may compete but the
team score is the sum of the best 3 or 5 top scores.  Each contest
consists of a written section and a programming section.
1.5.4 4.2.6 1.10.1 3.2.1 6.11.6 2.9.8 4.10.3 5.18.1
```

Output:

```
python 3
```

Explanation:

The first encoded string "1.5.4" tells you to find the 1st sentence, the 5th word in that sentence which is "computer", and then the 4th character in that word which is the lowercase letter "p". All others are done in a similar fashion. The string "4.10.3" is a space because there are only 9 words in the 4th sentence.

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INPUT: Input a string representing the text to be used in the decoding and another string representing the encoded message to be decoded. The first string will be no more than 2000 characters and the second string will be no longer than 200 characters.

OUTPUT: Output the decoded message as a string of characters.

SAMPLE INPUT:

1. ACSL is an international computer science competition among more than 300 schools. With countrywide and worldwide participants it became the American Computer Science League. It has been in continuous existence since 1978. Each yearly competition consists of 4 regular season contests. All students at each school may compete but the team score is the sum of the best 3 or 5 top scores. Each contest consists of a written section and a programming section.
   1.5.4 4.2.6 1.10.1 3.2.1 6.11.6 2.9.8 4.10.3 5.18.1
2. To be or not to be that is the question is a quote by William Shakespeare. 2B or not 2B is also a Boolean expression. Write it both ways.
   2.1.2 1.3.1 1.4.2 2.8.4 1.13.5 2.7.1 2.3.1 1.17.2 1.15.5 1.10.4 4.1.1 1.15.6 2.8.4 3.4.1 1.7.3 3.4.3 1.16.6 3.3.5 1.1.1 1.3.2 1.13.2 1.10.3
3. Programming languages are Java Python Visual BASIC Lisp FORTRAN C SQL. Javascript is the language of the Internet. HTML stands for Hypertext Markup Language and is not really a coding language. There are over 300 languages. Pick the one you like best.
   2.1.1 1.2.6 2.1.3 1.3.1 1.12.3 2.1.8 2.2.2 4.6.1 3.2.2 4.1.2 2.6.3 5.3.4 3.13.1 3.13.6 3.6.3 3.12.6 3.5.5 1.6.5 3.6.4 3.4.4 6.1.2 4.3.1 3.3.1 1.13.6 1.9.6 1.5.1 3.14.1 1.10.1 1.11.1
4. Four score and seven years ago our fathers brought forth on this continent a new nation conceived in Liberty and dedicated to the proposition that all men are created equal. Now we are engaged in a great civil war testing whether that nation or any nation so conceived and so dedicated can long endure. We are met on a great battlefield of that war. We have come to dedicate a portion of that field as a final resting place for those who here gave their lives that that nation might live.

It is altogether fitting and proper that we should do this.
This was written by Abraham Lincoln in November 1863.
2.1.1 1.2.3 8.6.3 6.9.1 1.31.1 1.8.7 5.6.1 1.13.7 1.15.2 2.18.1 2.12.2 8.10.4 6.6.5 4.8.2 7.4.1 1.16.2 3.7.5 4.10.4 3.12.1 1.21.7 2.5.1 6.5.7 1.13.7

5. There are 10 kinds of people in the world who are those who know binary and those who do not. Thus you should learn binary.
1.18.1 2.3.2 2.5.4 1.20.3 3.1.4 1.2.1 1.9.3 1.12.5 1.21.1 2.5.6 1.5.1 2.1.3

SAMPLE OUTPUT:

```
1. python 3
2. Boolean is always True
3. Java is the language of AP CS
4. No 1 speech of all time
5. what are you
```

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TEST DATA

TEST INPUT:

1. ACSL or the American Computer Science League is an international computer science competition among more than 300 schools. Originally founded in 1978 as the Rhode Island Computer Science League it then became the New England Computer Science League. With countrywide and worldwide participants it became the American Computer Science League. Each yearly competition consists of four regular season contests. All students at each school may compete but the team score is the sum of the best 3 or 5 top scores. Each contest consists of a written section and a programming section. Written topics tested include 12 different CS concepts.
   1.1.1 1.5.3 1.5.7 1.10.5 1.10.9 1.12.6 1.16.3 1.13.11 9.1.1 2.18.1 1.13.2 1.15.1 1.13.4 2.2.3 1.11.6 1.7.2 2.1.2 1.19.1 2.19.1 1.12.2 1.12.3 1.7.6 1.13.11 2.10.2 1.12.4 2.3.3 1.7.1 1.5.7 1.10.7 2.17.3 2.11.5 1.7.6 4.10.5 1.4.1 2.9.1 2.19.1 1.7.1 4.7.8 5.12.1 1.18.7 5.6.4 2.2.1 1.7.5 1.9.2
2. We hold these truths to be self evident that all men are created equal that they are endowed by their Creator with certain unalienable Rights that among these are Life Liberty and the pursuit of Happiness. That to secure these rights Governments are instituted among Men deriving their just powers from the consent of the governed.
   1.30.1 1.20.4 1.8.2 2.12.3 3.1.1 1.14.5 1.25.2 1.35.2 1.6.2 3.4.5 1.36.3 1.24.1 1.17.2 1.28.4 2.8.7 1.8.5 1.37.1 1.22.4 1.23.5 1.36.3 1.36.4 1.23.6 1.24.2 1.8.1 1.36.8 1.36.9 2.21.1 1.31.3 1.7.2 2.2.3 1.35.2 1.17.2 1.7.2 1.8.1
3. The upheavals of Artificial Intelligence can escalate quickly and become scarier and even cataclysmic. Imagine how a medical robot originally programmed to rid cancer could conclude that the best way to obliterate cancer is to exterminate humans who are genetically prone to the disease.
   1.4.1 2.1.1 3.5.2 2.6.3 1.2.9 1.15.6 2.2.1 2.7.9 1.11.7 2.30.7 2.31.5 1.7.7 1.3.1 3.2.1 1.2.9 1.14.3 2.3.1 2.16.3
4. I am increasingly inclined to think that there should be some regulatory oversight maybe at the national and international level just to make sure that we do not do something very foolish. I mean with artificial intelligence we are summoning

the demon. This was said by Elon Musk at the MIT AeroAstro Centennial Symposium.
3.10.1 2.1.1 4.1.2 1.4.1 3.12.6 1.33.3 1.8.5 1.31.1 3.10.2 1.4.6 2.11.1 3.3.1 2.4.7 3.7.1 2.7.2 3.4.2 2.11.7 1.32.1 2.10.4 3.10.8 1.1.2 1.1.1 3.1.1 3.9.4 2.5.8 3.11.5 1.17.1 3.1.3 1.12.4 3.6.3 2.5.12 1.11.1

5. Quadratic equations can be solved by factoring if the discriminant is a perfect square like 49 or 36. If it is not then there are two irrational roots. However if the discriminant is less than 0 the roots are 2 imaginary numbers. In the 1st two cases the parabola crosses the x axis in 1 or 2 places and in the 2nd case it does not.
4.7.5 5.4.3 1.2.9 1.14.2 1.2.3 3.4.10 2.9.3 3.1.6 1.1.4 1.19.5 3.14.3 2.2.1 3.7.4 1.14.3 4.3.2 2.1.3 1.16.1 4.7.8 3.4.4 5.10.1 2.9.1 4.8.7 4.24.4 4.13.1 1.18.2 1.16.2

TEST OUTPUT:

1. American Computer Science League ACSL is fun
2. Live life pursue happiness be free
3. AI is here to stay
4. AI is even scary for IT geniuses
5. b squared minus 4ac is 169

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问题：书加密法是一种发送者和接收者使用相同的文本对信息进行编码和解码的方法。发送者对信息进行编码时，使用 *"s.w.c"* 格式将要发送的信息中的每个字母数字字符替换为该字符在文本中的位置。*"s.w.c"* 格式代表文本中句子的编号(s)，句子中单词的编号(w)，单词中字符的编号(c)，所有编号均从 1 开始。接收者通过查找每个 *"s.w.c"* 字符串在文本中指定的字符来对编码信息进行解码。

在此程序中，将给定文本以及需解码的编码信息。这个文本包含字母数字字符、空格以及句号，限制条件如下：

- 每个单词只包含字母数字字符
- 每个单词之间用一个空格分隔
- 句子以句号结束，且句子之间用两个空格分隔

编码信息将包含多个 *"s.w.c"* 字符串，每个字符串之间用一个空格分隔。
通过查找文本中的字符来创建解码信息。如果 *"s.w.c"* 字符串并不指向文本中的某个字符，则在解码信息中使用一个空格代替。

示例：
输入：

```
ACSL is an international computer science competition among more than
300 schools.  With countrywide and worldwide participants it became
the American Computer Science League.  It has been in continuous
existence since 1978.  Each yearly competition consists of 4 regular
season contests.  All students at each school may compete but the
team score is the sum of the best 3 or 5 top scores.  Each contest
consists of a written section and a programming section.
1.5.4 4.2.6 1.10.1 3.2.1 6.11.6 2.9.8 4.10.3 5.18.1
```

输出：

```
python 3
```

解释说明：
第一个编码字符串 "1.5.4" 告诉你要先查找到第 1 句话，接着找到这句话的第 5 个单词 "computer"，最后找到这个单词里的第 4 个字符，即小写字母 'p'。所有其它字符的查找方式都与此类似。因为第 4 句话中只有 9 个单词，所以字符串 "4.10.3" 是一个空格。

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输入：输入一个字符串，代表解码时要使用的文本，再输入另一个字符串，代表需解码的编码信息。第一个字符串不超过 2000 个字符，第二个字符串不超过 200 个字符。

输出：以字符串的形式输出解码信息。

样本输入：

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2. To be or not to be that is the question is a quote by William Shakespeare. 2B or not 2B is also a Boolean expression. Write it both ways.
   2.1.2 1.3.1 1.4.2 2.8.4 1.13.5 2.7.1 2.3.1 1.17.2 1.15.5 1.10.4 4.1.1 1.15.6 2.8.4 3.4.1 1.7.3 3.4.3 1.16.6 3.3.5 1.1.1 1.3.2 1.13.2 1.10.3
3. Programming languages are Java Python Visual BASIC Lisp FORTRAN C SQL. Javascript is the language of the Internet. HTML stands for Hypertext Markup Language and is not really a coding language. There are over 300 languages. Pick the one you like best.
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4. Four score and seven years ago our fathers brought forth on this continent a new nation conceived in Liberty and dedicated to the proposition that all men are created equal. Now we are engaged in a great civil war testing whether that nation or any nation so conceived and so dedicated can long endure. We are met on a great battlefield of that war. We have come to dedicate a portion of that field as a final resting place for those who here gave their lives that that nation might live.

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5. There are 10 kinds of people in the world who are those who know binary and those who do not. Thus you should learn binary.
1.18.1 2.3.2 2.5.4 1.20.3 3.1.4 1.2.1 1.9.3 1.12.5 1.21.1 2.5.6 1.5.1 2.1.3

样本输出：

```
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测试数据

测试输入：

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测试输出：

1. American Computer Science League ACSL is fun
2. Live life pursue happiness be free
3. AI is here to stay
4. AI is even scary for IT geniuses
5. b squared minus 4ac is 169